

INTAKE-ARBITER -- FREE-COOLING DECISION REPORT

An agent that decides, hour by hour, whether outside air can cool a data centre.

THE SITE

Location Amazon Web Services, OR -- station KRLD, America/Los_Angeles
Committed pair OSM 460070145 -> 460070147
Amazon Web Services / Amazon Web Services
Facade gap 60.7 m between the two halls
Weather record 42,730 real hourly records from KRLD
Plume physics 576 steady-state solves on this site's own footprints; worst intake rise 0.5152 C at 60 deg

FortyGuard data: weather and geometry are this site's own; the four measured FortyGuard level offsets and the N-26 coverage record are Ashburn's, because only Ashburn has forecast/outcome day pairs. Quote the hours for this site; quote the coverage for Ashburn.

THIS REPORT IS A SNAPSHOT, NOT THE LIVE PAGE

The interface recomputes the decision for whatever configuration a reader selects. This file was generated at build time for the ONE configuration named below, chosen because it is the configuration in which this site's agent does the most while still changing mode at least once. If the numbers on screen differ from the numbers here, the configuration differs -- compare the two lists before concluding anything else.

Day 2024-06-02 (recirc_edge)
Plant limit 21.0 C
Notice required 3 h
Forecast skill 0.50 relative to persistence
Level anchor sensor
Condenser bank longest facade
Switch budget 2 mode changes per day
Minimum dwell 3 h
Max dew point 15.0 C (Green Grid WP#46 p.6)

WHAT THE AGENT DID ON THIS DAY

The agent ran free cooling for 7 of 24 hours with 2 mode changes. 3 hours were safe but still ran chillers, because the schedule could not afford them.

Agent 7 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
Incumbent 7 of 24 hours free cooling, 2 mode change(s), 0 unsafe hour(s)
3 hour(s) passed every gate and still ran chillers, because the schedule could not afford them

EVERY HOUR, AND THE REASON FOR IT

hh	mode	safe	binding	bound	limit	actual	margin
00	FREE	yes	-	18.441	21.0	18.080	1.532
01	FREE	yes	-	18.252	21.0	17.892	1.379
02	FREE	yes	-	18.416	21.0	17.910	1.315
03	FREE	yes	-	17.996	21.0	16.970	1.240
04	FREE	yes	-	18.496	21.0	17.800	1.181
05	FREE	yes	-	18.026	21.0	16.970	1.129
06	FREE	yes	-	18.577	21.0	17.783	1.393
07	mechanical	no	dry-bulb	21.660	21.0	19.780	2.003
08	mechanical	no	dry-bulb	22.972	21.0	20.781	2.326
09	mechanical	no	dry-bulb	24.942	21.0	22.780	2.159
10	mechanical	no	dry-bulb	25.836	21.0	22.780	1.941
11	mechanical	no	dry-bulb	26.088	21.0	23.330	1.701
12	mechanical	no	dry-bulb	26.575	21.0	23.221	1.566

13	mechanical	no	dry-bulb	25.971	21.0	23.000	1.428
14	mechanical	no	dry-bulb	24.103	21.0	19.721	1.324
15	mechanical	no	dry-bulb	23.293	21.0	18.332	1.331
16	mechanical	no	dry-bulb	21.925	21.0	18.013	1.276
17	mechanical	yes	-	19.530	21.0	17.152	1.372
18	mechanical	no	dew point	17.986	21.0	17.227	1.364
19	mechanical	no	dew point	18.094	21.0	18.331	1.648
20	mechanical	no	dew point	17.989	21.0	18.500	2.139
21	mechanical	no	dew point	17.550	21.0	17.780	2.188
22	mechanical	yes	switch budget	18.371	21.0	18.221	1.936
23	mechanical	yes	switch budget	18.419	21.0	18.000	1.607

bound = the agent's 90 %-nominal upper bound on intake air, forecast plus margin

actual = what the intake temperature turned out to be, from the station record

margin = group-conditional forecast error for that hour of day, plus plume spread

THE REASONING, HOUR BY HOUR

00:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.441 C, which is 2.559 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.532 C, and the margin is measured, not chosen: 1.281 C of group-conditional forecast error for this hour of day, plus 0.2507 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

01:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.252 C, which is 2.748 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.379 C, and the margin is measured, not chosen: 1.144 C of group-conditional forecast error for this hour of day, plus 0.2355 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

02:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.416 C, which is 2.584 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.315 C, and the margin is measured, not chosen: 1.064 C of group-conditional forecast error for this hour of day, plus 0.2507 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

03:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 17.996 C, which is 3.004 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.240 C, and the margin is measured, not chosen: 0.990 C of group-conditional forecast error for this hour of day, plus 0.2507 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

04:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.496 C, which is 2.504 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.181 C, and the margin is measured, not chosen: 0.930 C of group-conditional forecast error for this hour of day, plus 0.2507 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

05:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.026 C, which is 2.974 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.129 C, and the margin is measured, not chosen: 0.879 C of group-conditional forecast error for this hour of day, plus 0.2507 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

06:00 FREE-COOLING

Free cooling. The 90 %-nominal upper bound on intake air is 18.577 C, which is 2.423 C under the 21.0 C plant limit. That bound is the forecast plus a margin of 1.393 C, and the margin is measured, not chosen: 1.237 C of group-conditional forecast error for

this hour of day, plus 0.1553 C for how much the plume could move if the wind direction is off by the amount FortyGuard's forecast is actually off by.

07:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.659 C against a 21.0 C limit, so it fails by 0.659 C. It would take a limit 0.659 C higher, or a bound 0.659 C tighter, to change this hour.

08:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 22.972 C against a 21.0 C limit, so it fails by 1.972 C. It would take a limit 1.972 C higher, or a bound 1.972 C tighter, to change this hour.

09:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.942 C against a 21.0 C limit, so it fails by 3.942 C. It would take a limit 3.942 C higher, or a bound 3.942 C tighter, to change this hour.

10:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.836 C against a 21.0 C limit, so it fails by 4.836 C. It would take a limit 4.836 C higher, or a bound 4.836 C tighter, to change this hour.

11:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.088 C against a 21.0 C limit, so it fails by 5.088 C. It would take a limit 5.088 C higher, or a bound 5.088 C tighter, to change this hour.

12:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 26.575 C against a 21.0 C limit, so it fails by 5.575 C. It would take a limit 5.575 C higher, or a bound 5.575 C tighter, to change this hour.

13:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 25.970 C against a 21.0 C limit, so it fails by 4.970 C. It would take a limit 4.970 C higher, or a bound 4.970 C tighter, to change this hour.

14:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 24.103 C against a 21.0 C limit, so it fails by 3.103 C. It would take a limit 3.103 C higher, or a bound 3.103 C tighter, to change this hour.

15:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 23.293 C against a 21.0 C limit, so it fails by 2.293 C. It would take a limit 2.293 C higher, or a bound 2.293 C tighter, to change this hour.

16:00 MECHANICAL [dry-bulb]

Mechanical. The upper bound on intake air is 21.925 C against a 21.0 C limit, so it fails by 0.925 C. It would take a limit 0.925 C higher, or a bound 0.925 C tighter, to change this hour.

17:00 MECHANICAL

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 19.530 C against a 21.0 C limit. The schedule is what forbids it: no relaxation of either the switch budget or the dwell would free it, so the surrounding hours are worth more. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

18:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 17.986 C would have passed. The outside air is too HUMID: the dew-point bound is 15.41 C against a 15.0 C maximum, failing by 0.41 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

19:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 18.094 C would

have passed. The outside air is too HUMID: the dew-point bound is 16.05 C against a 15.0 C maximum, failing by 1.05 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

20:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 17.989 C would have passed. The outside air is too HUMID: the dew-point bound is 16.34 C against a 15.0 C maximum, failing by 1.34 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

21:00 MECHANICAL [dew point]

Mechanical, and TEMPERATURE IS NOT THE REASON -- the dry-bulb bound of 17.550 C would have passed. The outside air is too HUMID: the dew-point bound is 16.05 C against a 15.0 C maximum, failing by 1.05 C. Cool but damp air condenses on cold surfaces inside the hall, which is why real economizers gate on humidity and not on temperature alone.

22:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.371 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

23:00 MECHANICAL [switch budget]

Mechanical EVEN THOUGH THIS HOUR IS SAFE. Every gate passes -- the bound is 18.419 C against a 21.0 C limit. The schedule is what forbids it: the switch budget of 2 changes per day is already committed elsewhere. This is the one explanation a thermostat cannot give, because a thermostat has no plan to be constrained by. It is also why the decision is a schedule and not a comparison: spending a mode change here would cost a better one later.

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